ENZYMES

Learning objectives

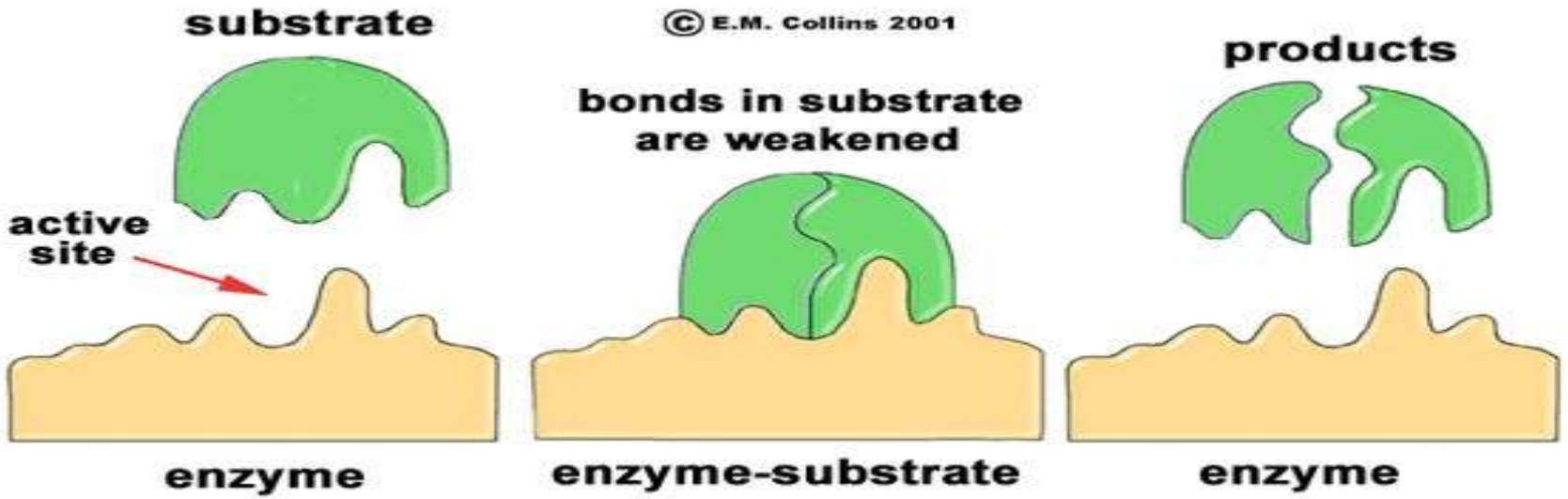
At the end of this session students will be able to understand;

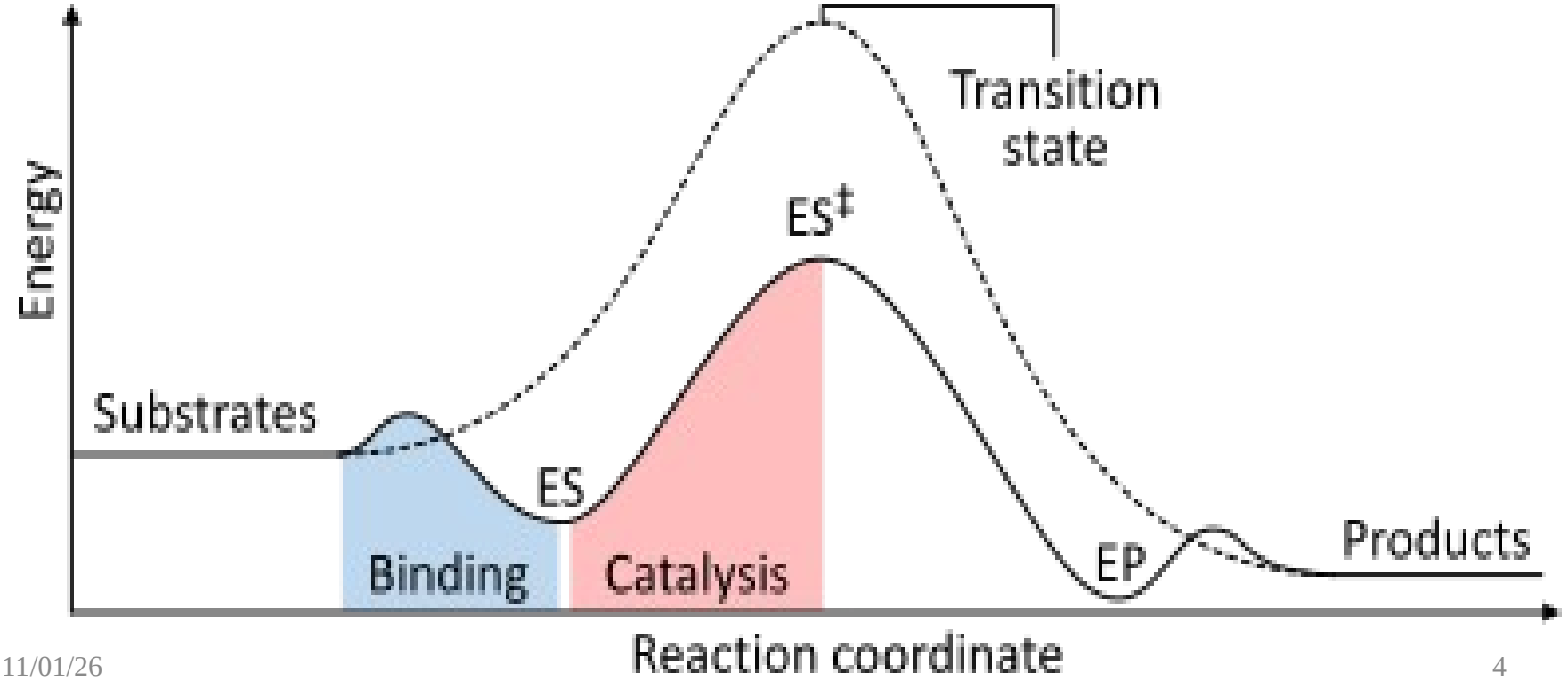
1.Enzyme and its classification.

2.Basic concepts involved in the biochemistry of enzyme with mechanisms.

3.Properties, inhibition, types and diagnostic importance of enzymes.

- Protein catalyst that increases the rate of reactions without themselves being changed in the overall process.
- Enzymes thus direct all metabolic events.
- Enzymes works selectively on the channel of reactants – Substrates.


substrate
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products
bonds in substrate
are weakened
active
site
enzyme
enzyme-substrate
enzyme


Energy
Transition
state
ES‡
Substrates
ES
EP
Products
Binding
Catalysis
Reaction coordinate

Nomenclature

- Each enzymes is assigned two names, the first is its short, Recommended name, convenient for everyday use.
- The second name is the more complex and considered to be complete- systematic name (used when the enzymes must be identified without ambiguity).
- For e.g., the pattern use generally is the suffix ***"ase"*** attached to the substrate of the reaction for example, glucosidase, urease, sucrase.
- In some cases a description of the action performed, for example lactate dehydrogenase.

Classification of Enzymes

- The IUBMB developed a system of nomenclature in which enzymes are divided into six major classes.

Table 5.1 Enzyme Classification Based on Reaction Types

Class	Type of Reaction Catalyzed	Example
Hydrolase	Hydrolysis (catabolic)	Lipase—breaks down lipid molecules
Isomerase	Rearrangement of atoms within a molecule (neither catabolic nor anabolic)	Phosphoglucoisomerase—converts glucose 6-phosphate into fructose 6-phosphate during glycolysis
Ligase or polymerase	Joining two or more chemicals together (anabolic)	Acetyl-CoA synthetase—combines acetate and coenzyme A to form acetyl-CoA for the Krebs cycle
Lyase	Splitting a chemical into smaller parts without using water (catabolic)	Fructose 1,6-bisphosphate aldolase—splits fructose 1,6-bisphosphate into G3P and DHAP
Oxidoreductase	Transfer of electrons or hydrogen atoms from one molecule to another	Lactic acid dehydrogenase—oxidizes lactic acid to form pyruvic acid during fermentation
Transferase	Moving a functional group from one molecule to another (may be anabolic)	Hexokinase—transfers phosphate from ATP to glucose in the first step of glycolysis

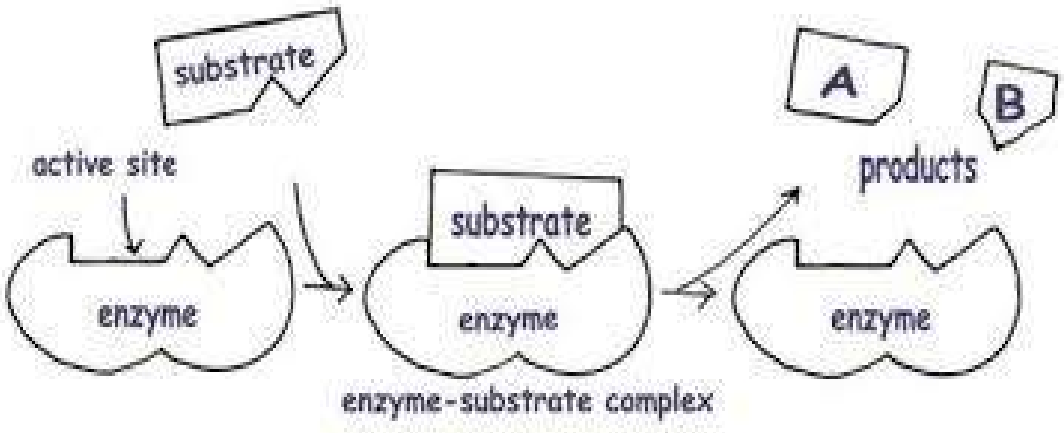
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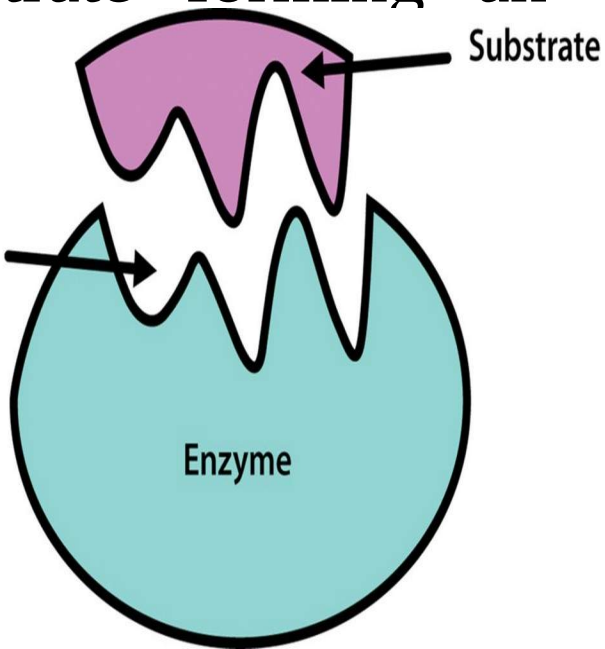
Properties of Enzymes

- They are protein catalyst that increases the velocity of a chemical reaction they catalyze.

Active sites

- Enzymes molecules contain special pocket or cleft called *active site*.
- Active site contains amino acids chains that create a three dimensional surface complementary to the substrate.
- The active site binds to the substrate forming an *Enzyme-Substrate* (ES) complex.





Catalytic efficiency

- Most enzyme- catalyzed reactions are highly efficient, proceeding from 10^3 to 10^8 times faster than un-catalyzed reactions.
- Typically, each enzyme molecule is capable of transforming 100 to 1000 substrate molecules into product each second.

Turnover number

- The number of molecules of substrate converted to product per enzyme molecule per second is called turnover number.

Specificity

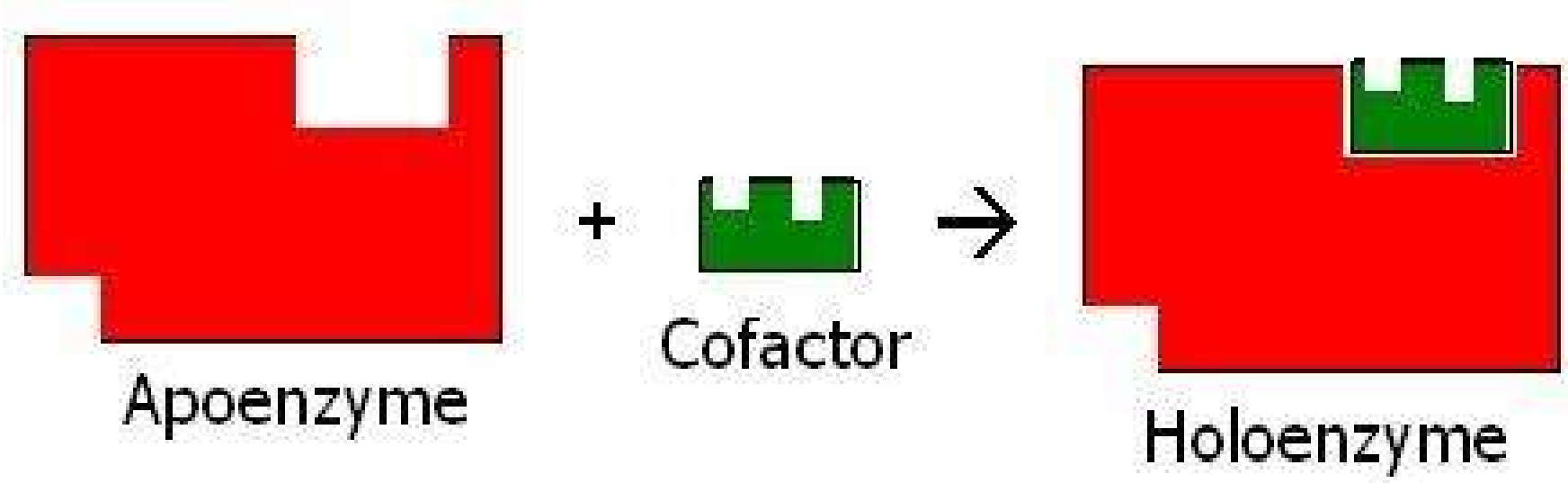
- Enzymes are highly specific, interacting with one or a few specific substrates and catalyzing only one type of chemical reactions.

Cofactors.

- Some enzymes associate with a non protein cofactor that is needed for enzyme activity.
- Commonly encountered cofactors include metal ions (Zn2+, Fe+2) and organic molecules, known as *coenzymes*.
- Cofactors can be metals or coenzymes, and their primary function is to assist in enzyme activity.
- ***Coenzymes*** (non-protein organic molecules) that are mostly derivatives of vitamins soluble in water by phosphorylation.
- They bind ***apoenzyme*** to proteins to produce an active ***holoenzyme***.

Holoenzyme

- The enzyme with its cofactor.
- A holoenzyme refers to a catalytically active enzyme that consists of both apoenzyme (enzyme without its cofactor(s)) and cofactor.

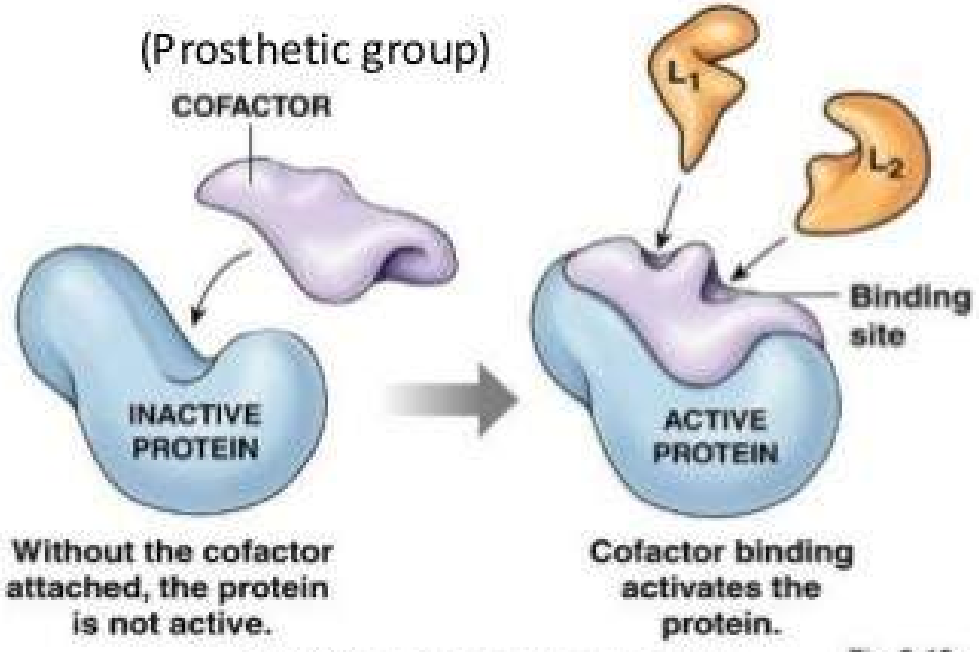


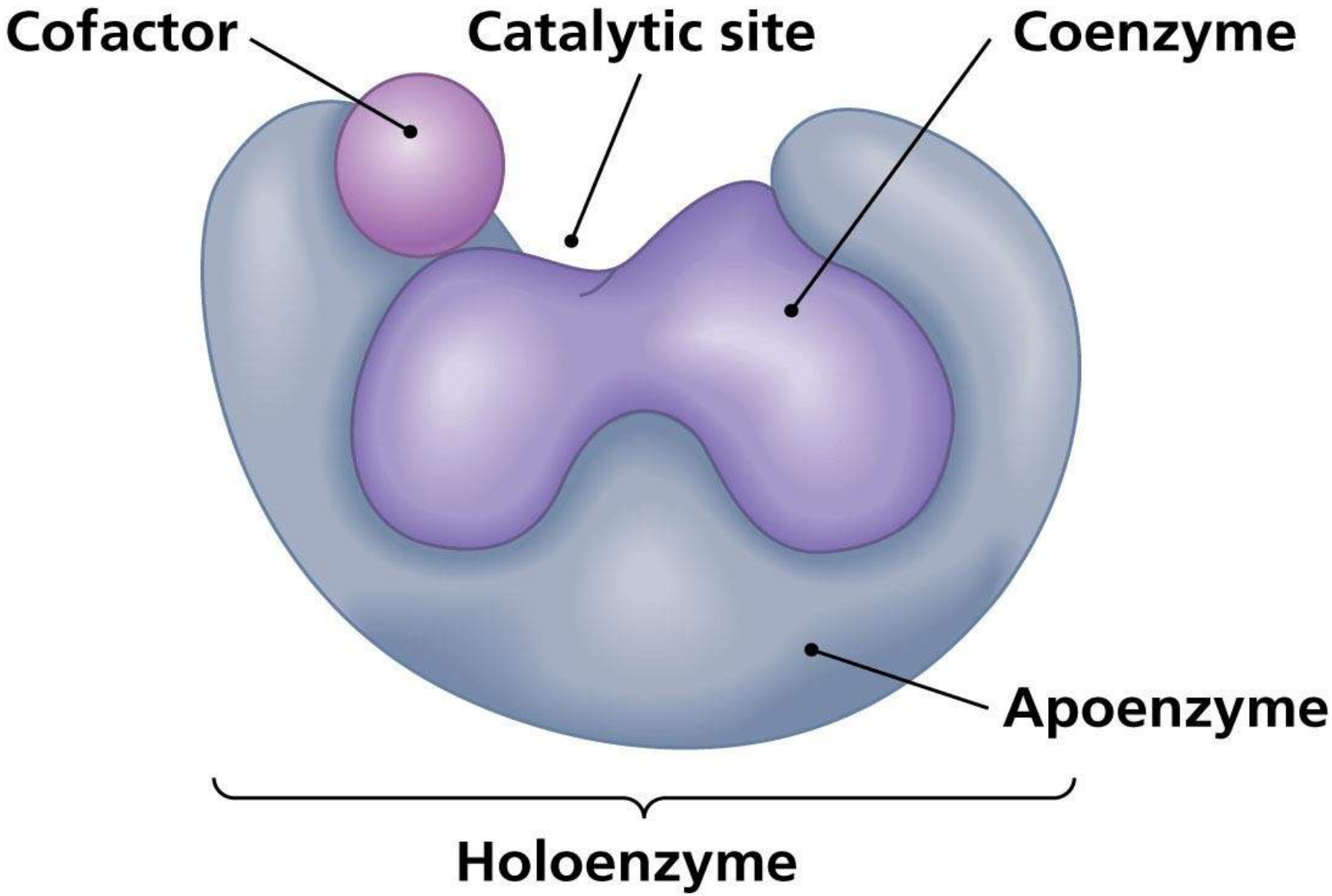
Apoenzyme

- The protein portion of the holoenzyme.
- In the absence of the appropriate cofactor, the apoenzyme typically does not show biologic activity.

Prosthetic group

- Is a tightly bound coenzyme that does not dissociate from the enzyme.
- In peroxidase and catalase, which catalyze the breakdown of hydrogen peroxide to water and oxygen, heme is the prosthetic group.





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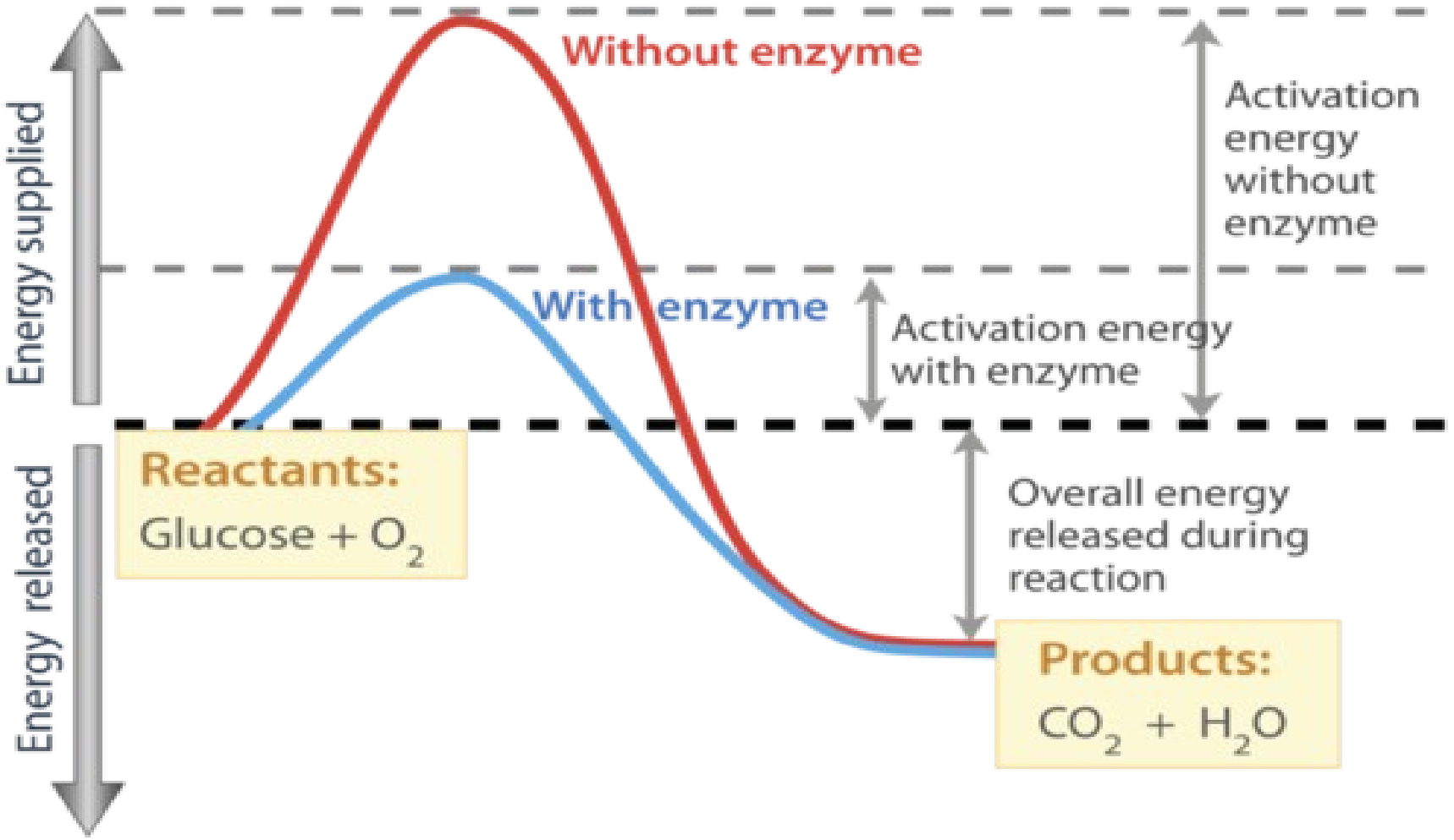
Regulation

- Enzyme activity can be regulated- that is, enzymes can be activated or inhibited so that the rate of product formation respond to the needs of the cells.

Enzyme Activity

- All chemical reactions have an energy barrier separating the reactants and the products.
- This barrier, called the free energy of activation.
- Is the energy difference between that of the reactants and a high energy intermediate that occurs during the formation of products.
- The lower the free energy of activation, the more molecules have sufficient energy to pass through the transition state, and, thus the faster the rate of reaction.

Enzyme Action



Alternate reaction pathway:

- The enzyme does not change the free energies of the reactant or products and therefore, does not change the equilibrium of the reaction.
- Providing an alternate reaction pathway with lower free energy of activation.
- Accelerate the rate of reaction with which equilibrium is reached.

Factors Affecting Reaction Velocity

Substrate concentration:

- The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per units time and is usually expressed as µmoles product formed per minute.
- Maximal velocity as the leveling off of the reaction rate at high substrate concentrations reflects the saturation with substrate of all available binding sites on the enzyme.

Hyperbolic shape of velocity *Vs* substrate

- Enzymes show the pattern of Michalis Menten kinetics, in which the graph shows the relation of Initial reaction velocity (Vo) against substrate concentration [S] is a hyperbolic relation.

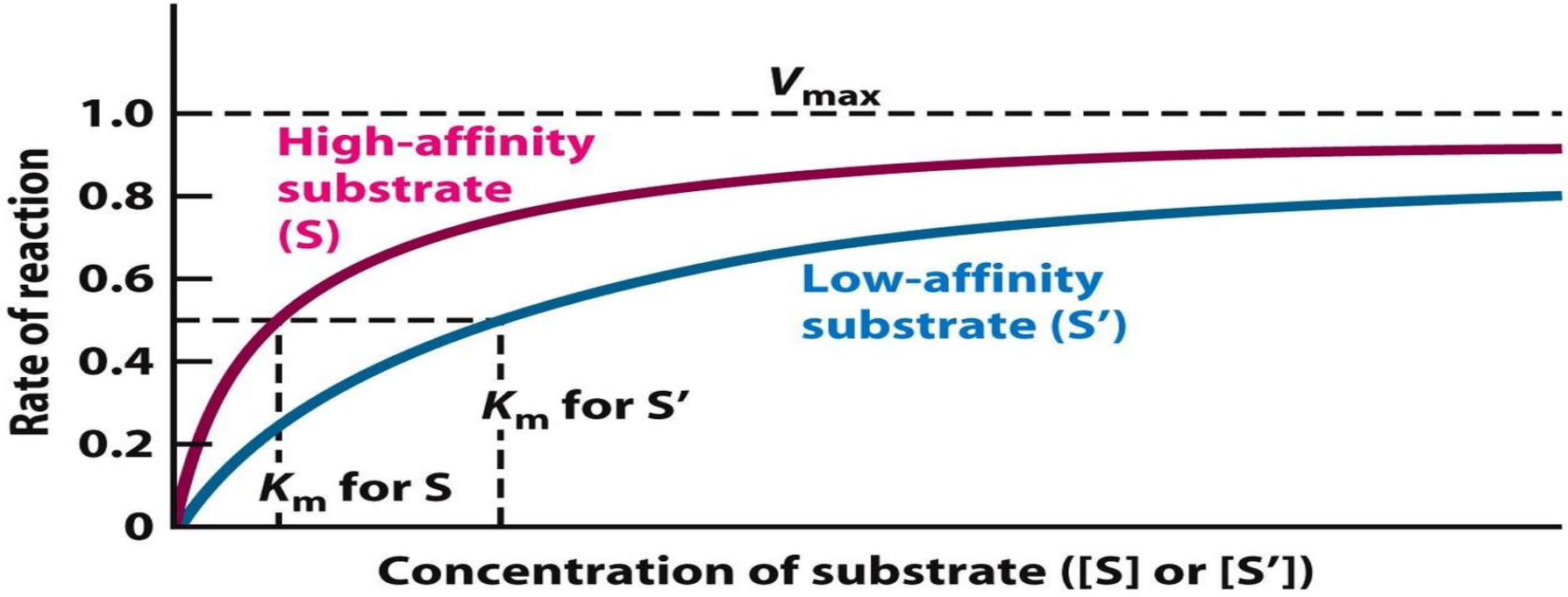
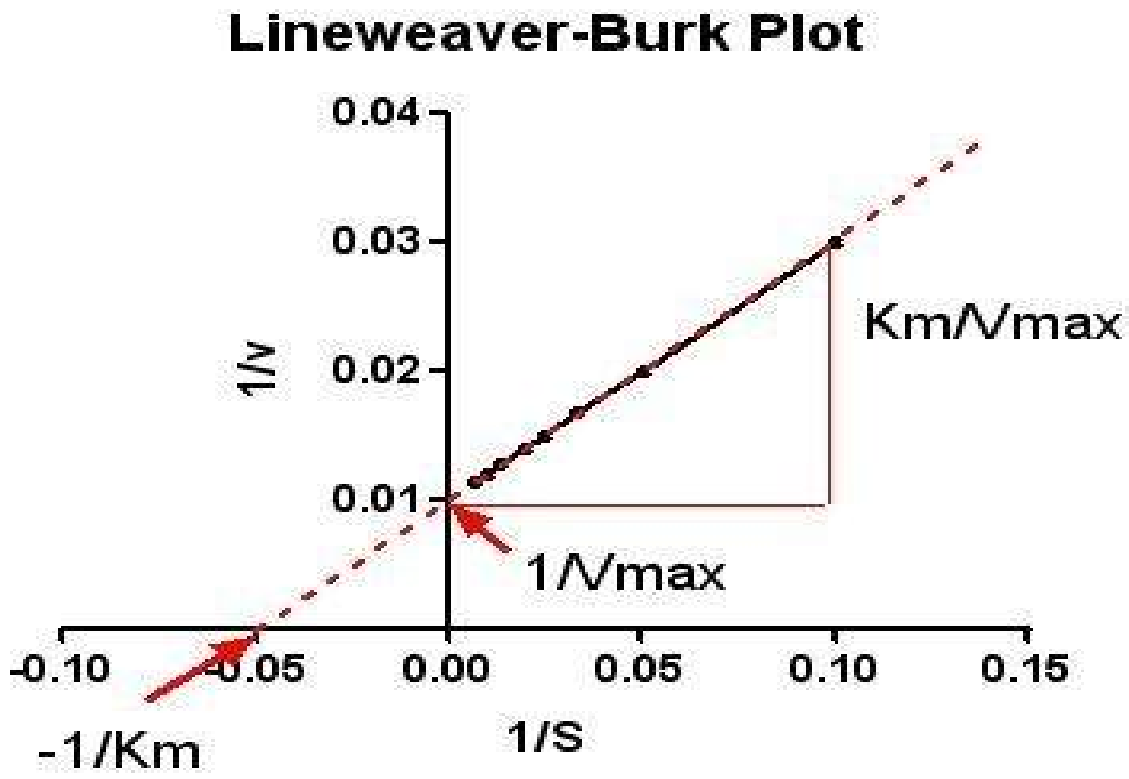


Figure 3-22b
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Lineweaver-Burk Plot

- The **Lineweaver–Burk plot** is a graphical representation of the Lineweaver–Burk equation of enzyme kinetics, described by Hans Lineweaver and Dean Burk in 1934.

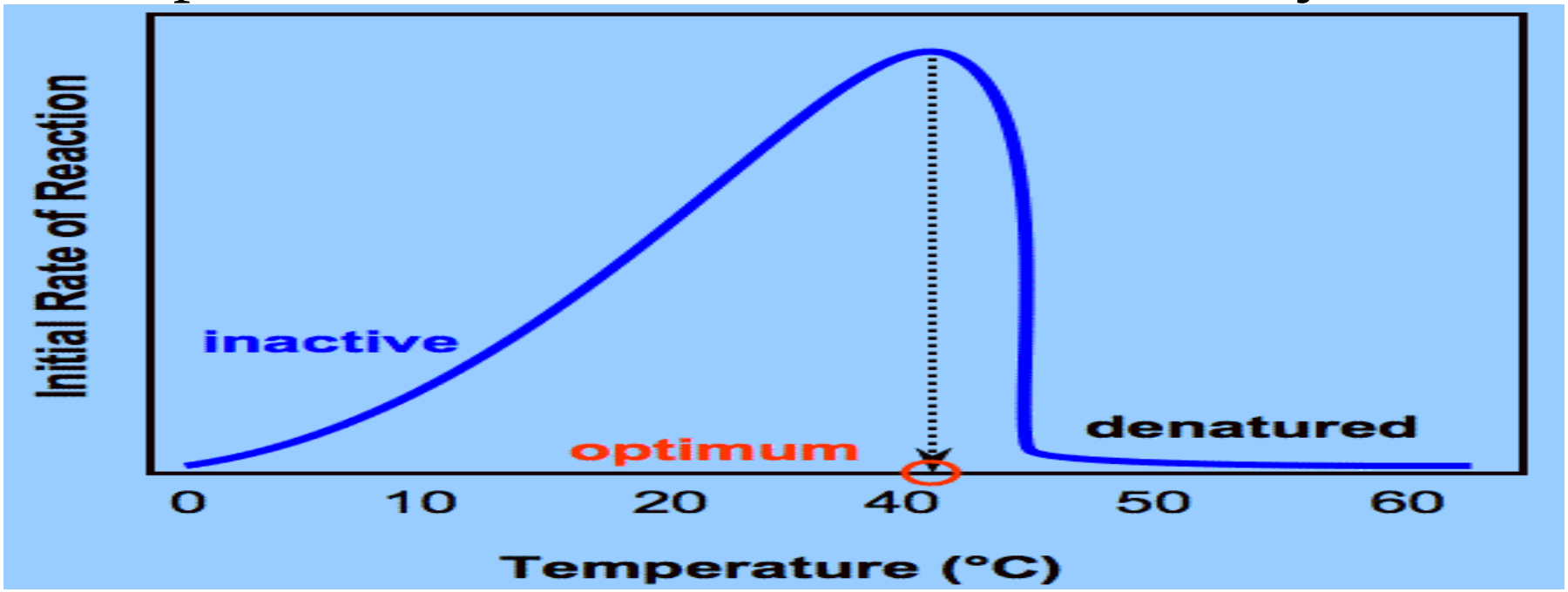
- The Lineweaver–Burk plot was widely used to determine important terms in enzyme kinetics, such as K_m and V_{max}, before the wide availability of powerful computers and non-linear regression software. The y-intercept of such a graph is equivalent to the inverse of V_{max}; the x-intercept of the graph represents $-1/K_m$. It also gives a quick, visual impression of the different forms of enzyme inhibition.
- The double reciprocal plot distorts the error structure of the data, and it is therefore unreliable for the determination of enzyme kinetic parameters.



The Lineweaver-Burk Plot creates a straight line graph that allows Vmax and Km to be read off the graph as -1/Km and 1/Vmax. (To convert these back to Vmax and Km, just calculate 1/(1/Vmax) or 1/(-1/Km)).You cannot calculate 1/0, so ignore this reading. Km/Vmax can also be read directly off the graph as the gradient (To convert into Vmax/Km, do 1/(Km/vmax)).

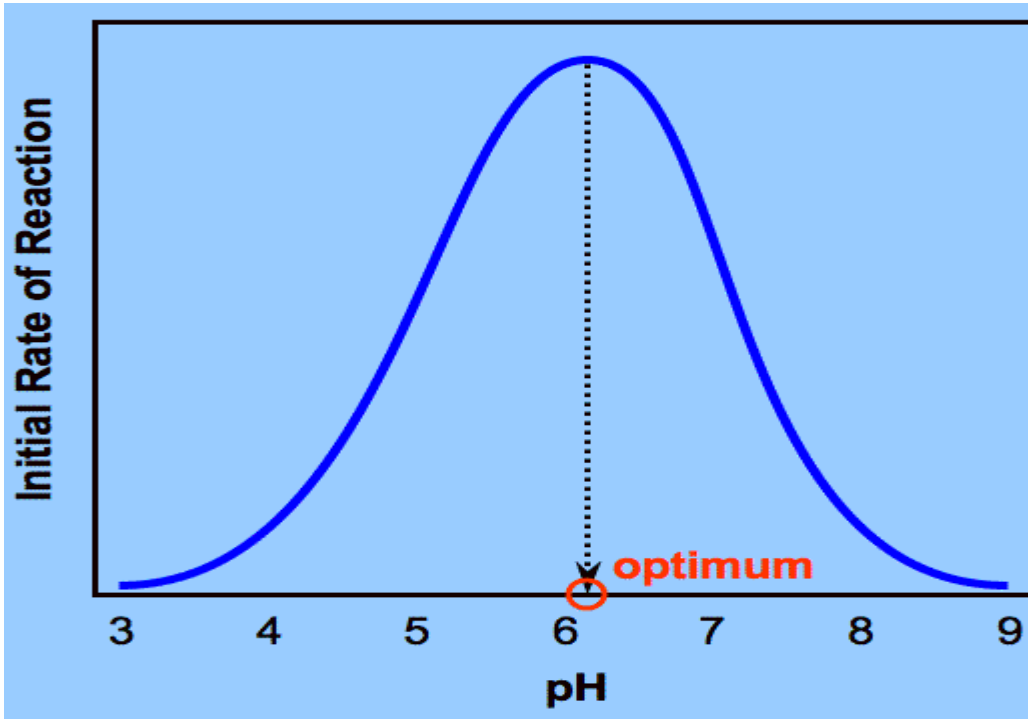
Temperature

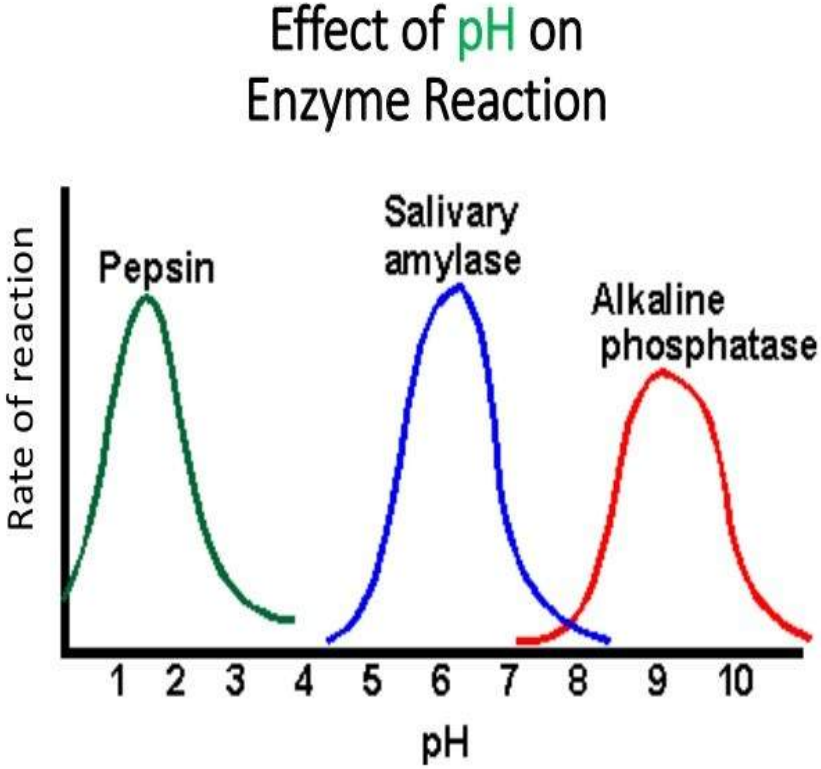
- The reaction velocity increases with temperature until a peak velocity is reached. This is the increase of velocity with temperature.
- Further elevation of the temperature results in a decrease in reaction velocity as a result of temperature induced denaturation of the enzyme.



pH

Effect of pH on ionization of active site: the catalytic process usually requires that the enzyme and substrate have specific chemical groups in an ionized or unionized state in order to interact.





- Extremes of pH can also leads to denaturation of the enzyme, because the structure of the catalytically active protein molecule depends on the ionic character of the amino acid side chain.
- The pH at which maximal enzyme activity is achieved- the ***pH optimum.***

Michaelis-Menten Kinetics Equation



S = substrate P = product

E = enzyme

ES = enzyme-substrate complex

k_1, k_{-1}, k_2, k_{-2} are rate constants

Michaelis-Menten Equation

$$v_o = \frac{[V_{max}][S]}{Km + [S]}$$

v_o = initial reaction velocity

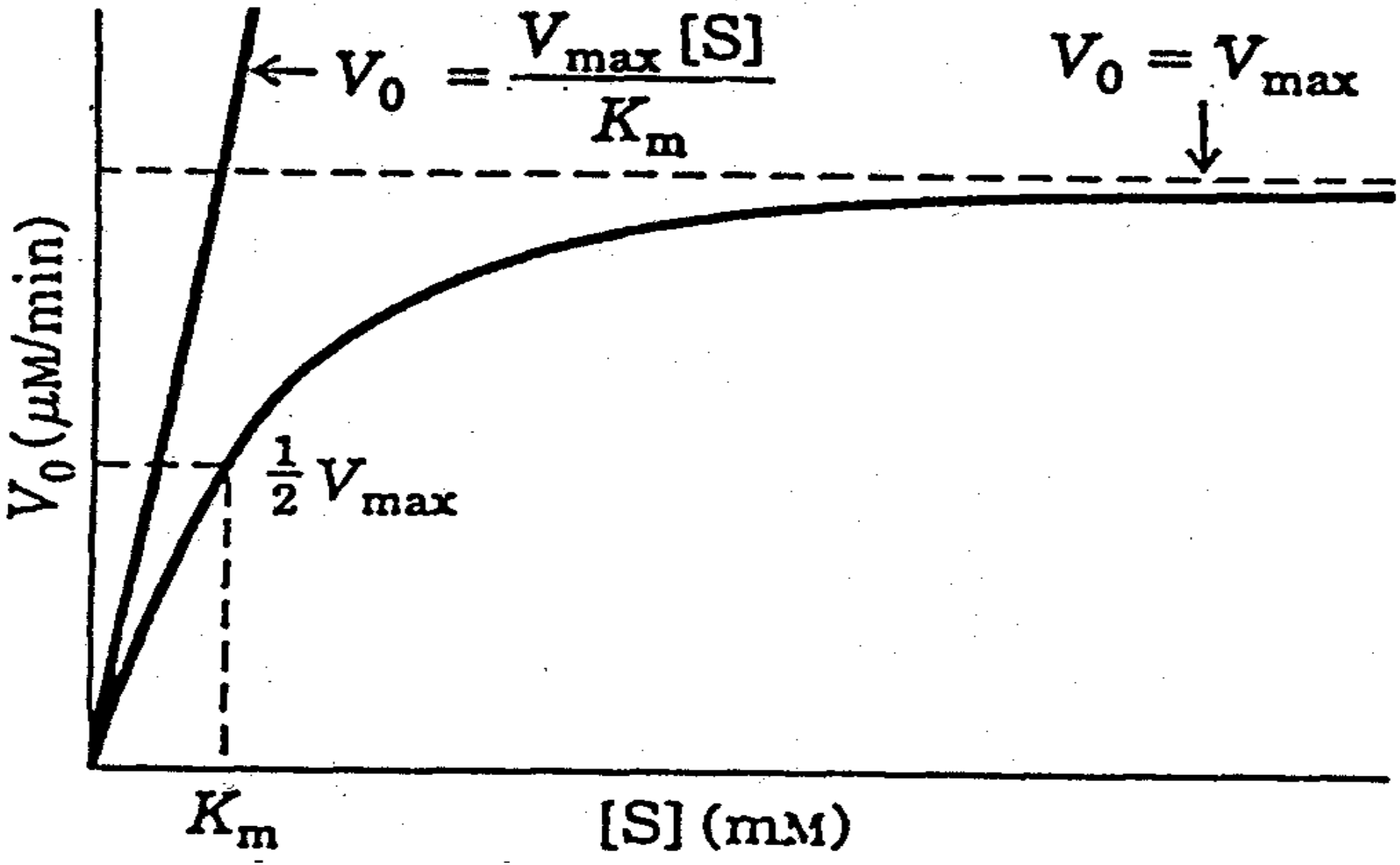
V_{max} = maximal velocity

[S] = substrate concentration

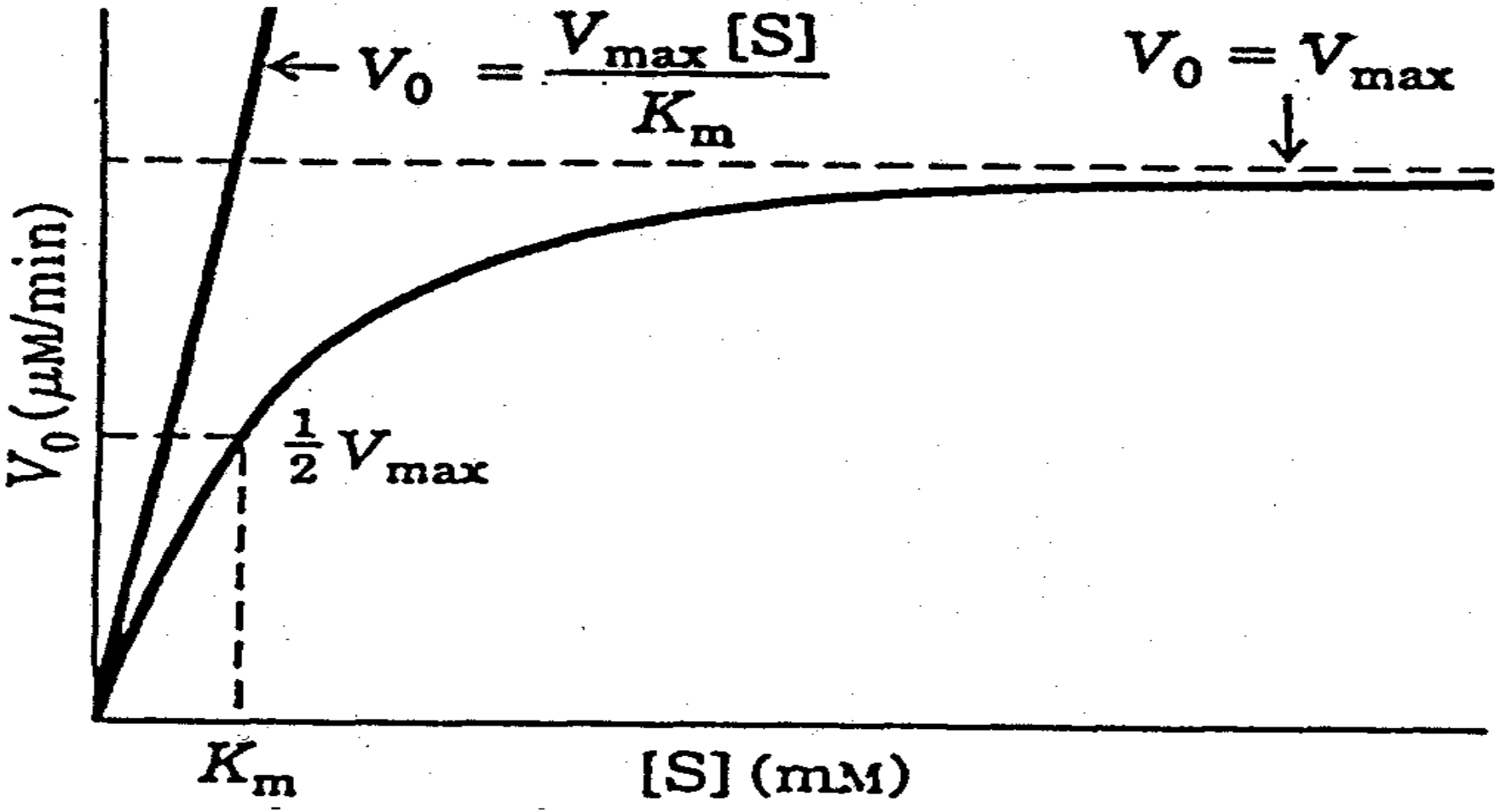
Initial Velocity (V_o) and [S]

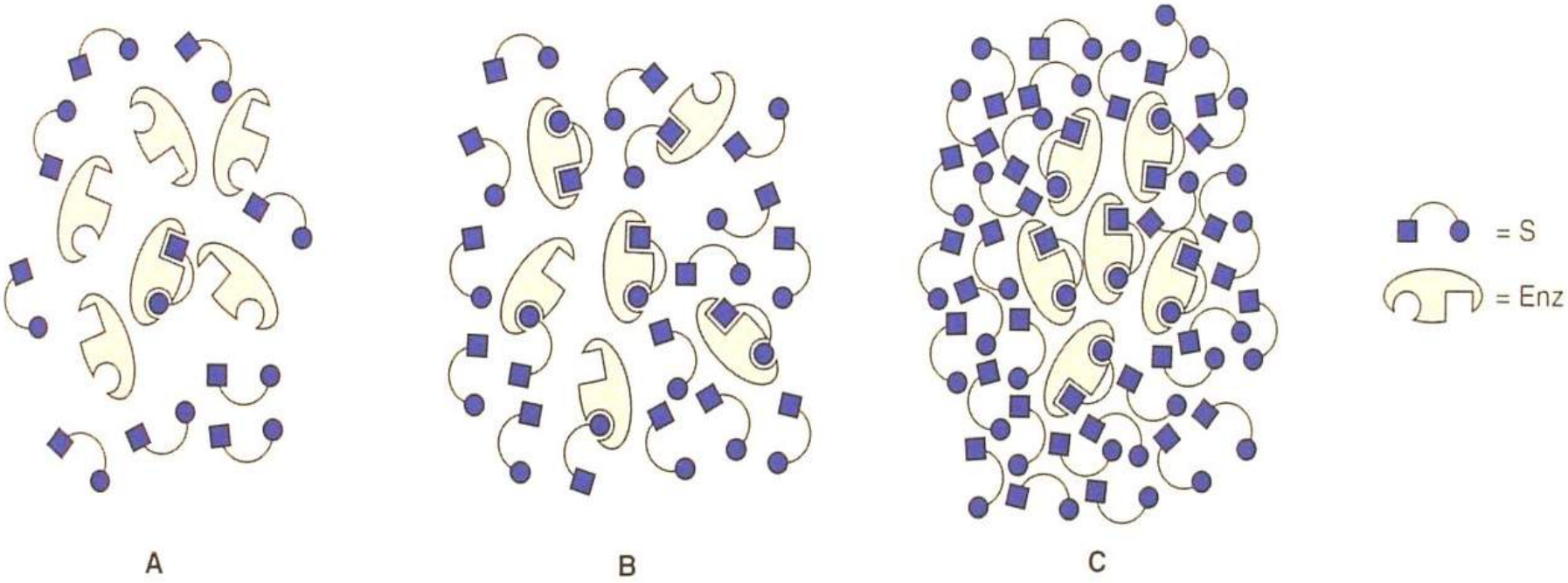
- The concentration of substrate [S] present greatly influence the rate of product formation, termed the velocity (v) of a reaction.
- Studying the effects of [S] on the velocity of a reaction is complicated by the reversibility of enzyme reactions, e.g. conversion of product back to substrate.
- To overcome this problem, the use of **initial velocity** (V_o) measurements are used.
- At the start of a reaction, [S] is in large excess of [P], thus the initial velocity of the reaction will be dependent on substrate concentration

Michaelis-Menten Curve



Michaelis-Menten Curve





A. Low [S] B. 50% [S] or Km C. High, saturating [S]

Steady State Assumption

- The M-M equation was derived in part by making several assumptions. An important one was: **the concentration of substrate must be much greater than the enzyme concentration.**
- **I**n the situation where [S] >> [E] and at initial velocity rates, it is assumed that the changes in the concentration of the intermediate ES complex are very small over time (v_o).
- This condition is termed a **steady-state rate,** and is referred to as **steady-state kinetics. Therefore, it follows that the rate of ES formation will be equal to the rate ES breakdown.**

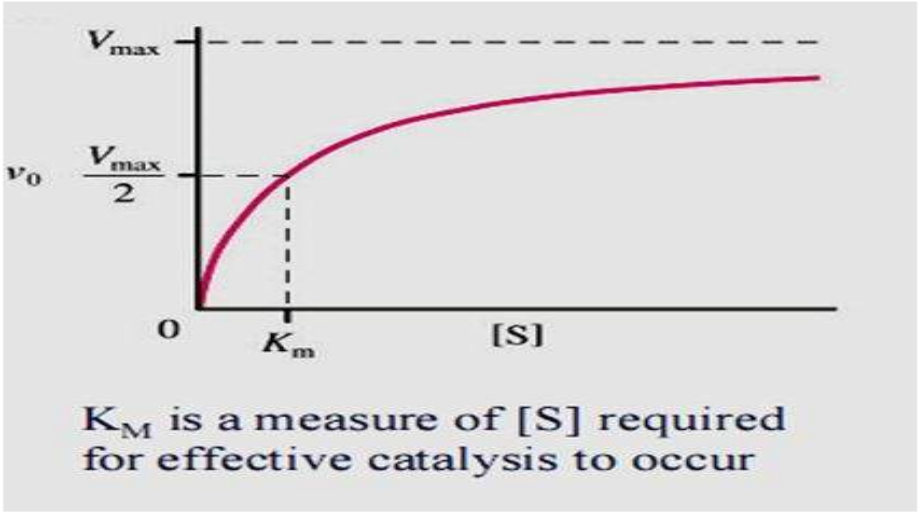
Meaning of K_m

- This means that at one half of the maximal velocity, the substrate concentration at this velocity will be equal to the K_m.

What does K_m mean?

K_m (the Michaelis constant) is a measure of the affinity of the enzyme for the substrate.

Substrates that bind tightly have small K_m values, substrates that bind weakly have large K_m values.



K_M is a measure of [S] required for effective catalysis to occur

- This relationship has been shown experimentally to be valid for many enzymes much more complex in regards to the number of substrates and catalytic steps than the simple single substrate model used to derive it.
- The significance of K_m will change based on the different rate constants and which step is the slowest (also called the rate-limiting step).
- In the simplest assumption, the rate of ES breakdown to product (k_2) is the rate-determining step of the reaction, so $k_{-1} >> k_2$ and $K_m = k_{-1}/k_1$.

Uses of K_m

- Experimentally, K_m is a useful parameter for characterizing the number and/or types of substrates that a particular enzyme will utilize (an example will be discussed).
- It is also useful for comparing similar enzymes from different tissues or different organisms.
- Also, it is the K_m of the rate-limiting enzyme in many of the biochemical metabolic pathways that determines the amount of product and overall regulation of a given pathway.

- Clinically, K_m comparisons are useful for evaluating the effects mutations have on protein function for some inherited genetic diseases.

Isozymes

- Multiple forms of enzymes.
- Enzymes that differ in amino acid sequence but catalyze the same chemical reaction.
- Synthesized from different genes.

Type	Composition	Location
LDH_1	HHHH	Heart and Erythrocyte
LDH_2	HHHM	Heart and Erythrocyte
LDH_3	HHMM	Brain and Kidney
LDH_4	HMMM	Skeletal Muscle and Liver
LDH_5	MMMM	Skeletal Muscle and Liver

Proenzymes

- Also called Zymogen.
- An inactive precursor of an enzyme.
- A zymogen requires a biochemical change (such as a hydrolysis reaction revealing the active site, or changing the configuration to reveal the active site) for it to become an active enzyme.
- Enzymes like pepsin are created in the form of pepsinogen, an inactive zymogen.



Enzyme Inhibition

- An enzyme inhibitor is a molecule that binds to an enzyme and decreases its activity.

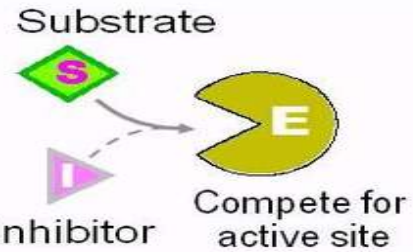
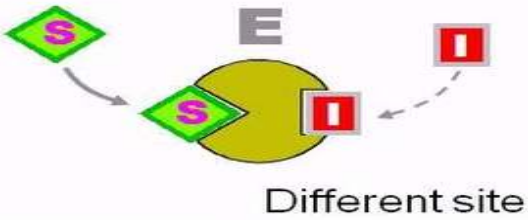
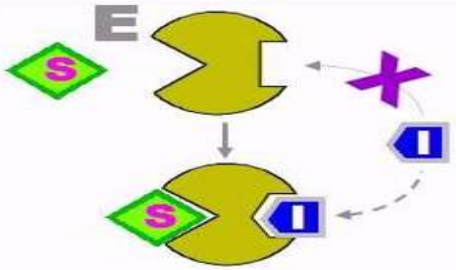
Types of Enzyme Inhibition

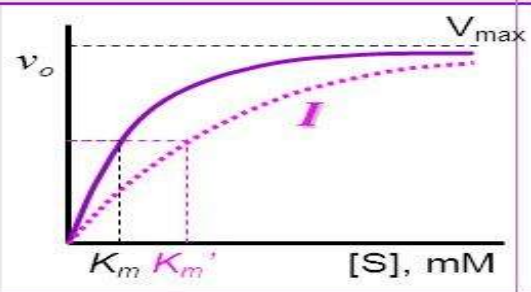
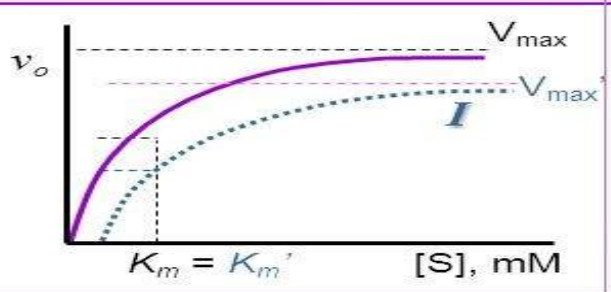
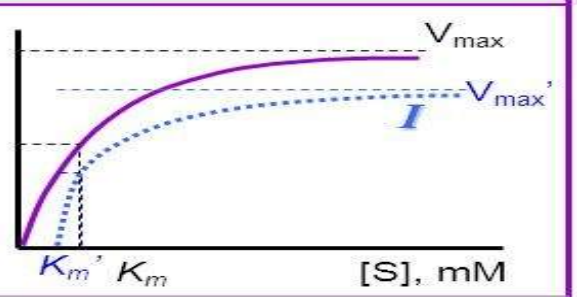
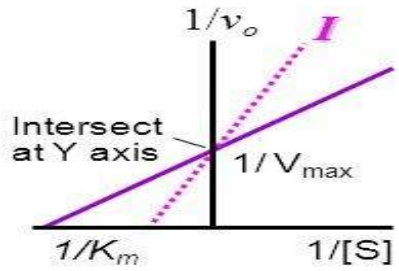
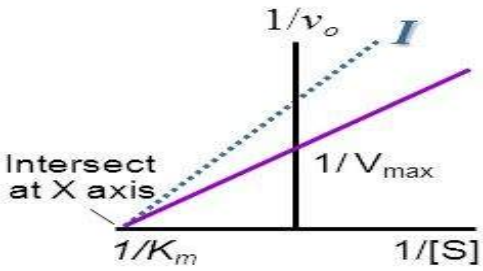
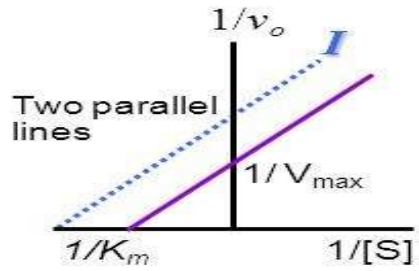
1. *Competitive inhibition*, the substrate and inhibitor cannot bind to the enzyme at the same time.
2. *Uncompetitive inhibition*, the inhibitor binds only to the substrate-enzyme complex.
3. *Non-competitive inhibition*, the binding of the inhibitor to the enzyme reduces its activity but does not affect the binding of substrate.

Enzyme Inhibitors



Enzyme Inhibition (Mechanism)

	Competitive	Non-competitive	Uncompetitive
Cartoon Guide			
Equation and Description	$E + S \rightleftharpoons ES \rightarrow E + P$ $+$ I $\updownarrow$ EI	$E + S \rightleftharpoons ES \rightarrow E + P$ $+$ $+$ I I $\updownarrow$ $\downarrow\uparrow$ $EI + S \rightarrow EIS$	$E + S \rightleftharpoons ES \rightarrow E + P$ $+$ I $\updownarrow$ EIS
	[*I*] binds to free [E] only, and competes with [S]; increasing [S] overcomes Inhibition by [*I*].	[*I*] binds to free [E] or [ES] complex; Increasing [S] can not overcome [*I*] inhibition.	[*I*] binds to [ES] complex only, increasing [S] favors the inhibition by [*I*].

	Competitive	Non-competitive	Uncompetitive
Direct Plots			
	V_{max} unchanged K_m increased	V_{max} decreased K_m unchanged	Both V_{max} & K_m decreased
Double Reciprocal			

Kompetitive inhibitor

Km increases





"competitive scissors"

Very efficient

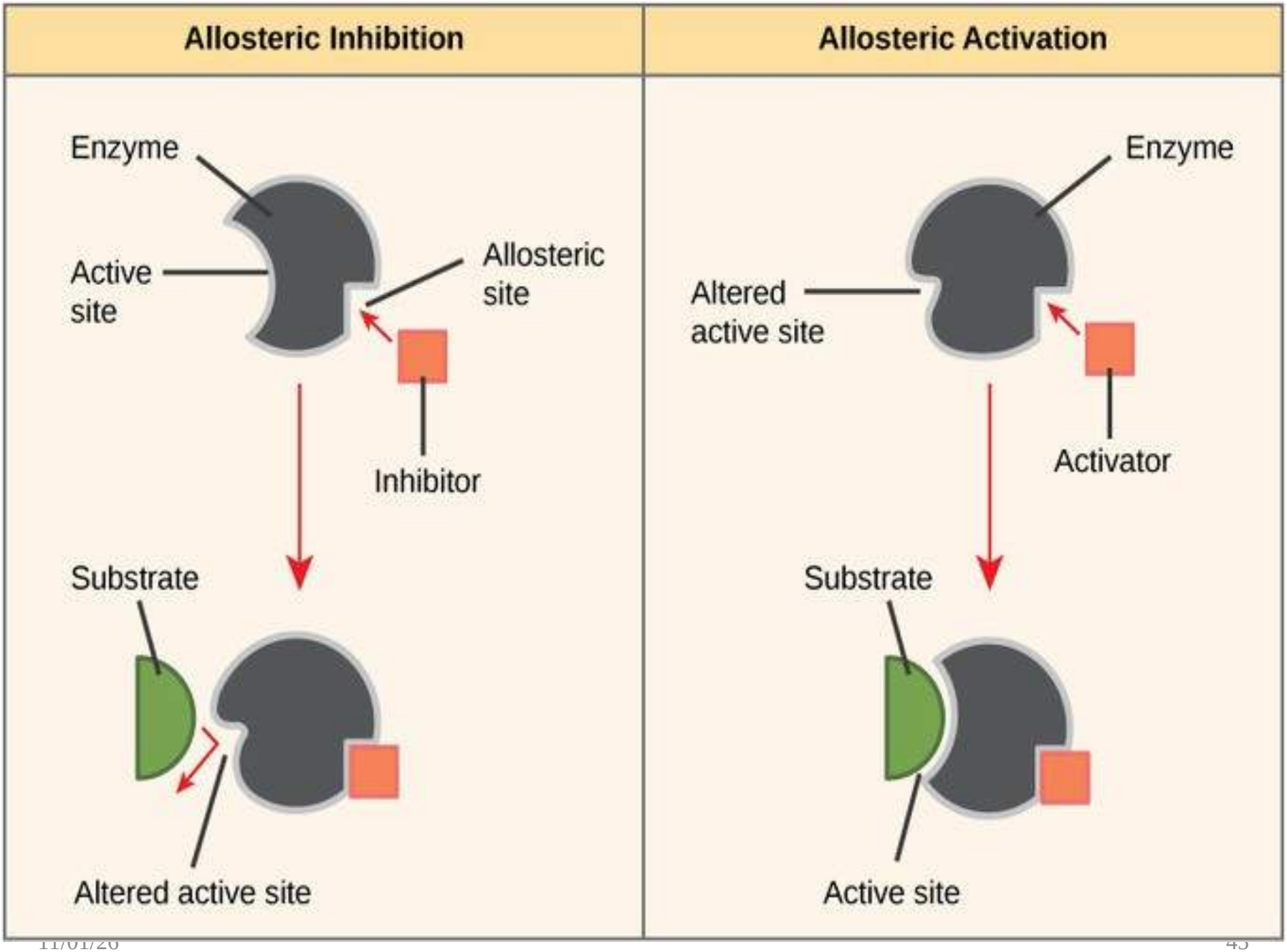
Vmax efficacy (Y axis)

Competent

Km potency (X axis)

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Competitive vs Non-Competitive Inhibitors mnemonic

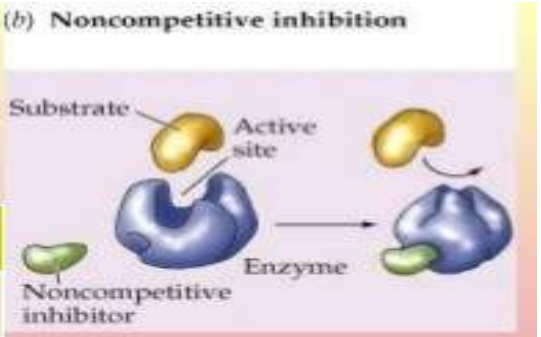

Allosteric Inhibition
Enzyme
Active site
Allosteric site
Inhibitor
Substrate
Altered active site
Allosteric Activation
Enzyme
Altered active site
Activator
Substrate
Active site

Clinically useful Competitive Inhibition

Drugs	Target Enzyme	Therapeutic Use
STATINS - Atorvastatin , simvastatin	HMG CoA reductase	Decrease plasma Cholesterol level - Antihyperlipidemic agents
Allopurinol	Xanthine oxidase	Gout
Methotrexate	Dihydrofolate reductase	Cancer
Captopril & Enalapril	Angiotensin converting enzyme	High blood pressure
Dicoumarol	Vit.K-epoxide-reductase	Anti-coagulant

Non competitive inhibitor

Inhibitor	Enzyme inhibited
Heavy metals - Ag^{2+} ,Hg^{2+} , Pb^{2+}	Binding with cysteinyl SH gr of E
Pepstatin	Pepsin
Soyabean trypsin inhibitor	Trypsin
Ethanol or narcotic drugs	Acid phosphatase



Measurements of enzymes

- in serum
- within a tissue
- enzymes as markers of disease,
- Injury to tissue releases cellular substances that can be used as plasma markers of tissue damage.
- enzyme release is highly specific for cell death in some settings.

ENZYMES IN DISEASES

Organ	Enzymes	Diseases
Liver	ALP,GDH,SGOT,SGPT,Glutamyl Tranferasess,Glutathione S Tranferases,	Viral/toxic hepatitis or cirrhosis, myocardial infraction,jaundice,
Heart	Creatne Kinases,LDH, AST/ALT,	Myocardial Infraction,skelital muscle disorders.
Other diseases	Amylases,Aldolases,TG Lipases,CK,ALP,ACP,Choline Esterases,HMG CO A Reductases,Xanthine Oxidases,	GIT disorders, jaundice,bone diseases, rickets,

Table 3. Enzyme markers of clinical significance

Enzyme (abbreviation)	Clinical significance
Acid phosphatase (ACP)	Prostatic carcinoma
Alkaline phosphatase (ALP)	Obstructive liver diseases, bone disorders
Alanine transaminase (ALT,GPT)	Hepatic disorders, viral hepatitis
Aspartate transaminase (AST,GOT)	Myocardial infarction, hepatic disorders
Alpha-amylase (AMS)	Acute pancreatitis
Aldolase (ALS)	Skeletal muscle disorders
Creatine kinase (CK)	Myocardial infarction, muscle disorders
Gamma-glutamyl transferase (GGT)	Hepatic disorders
G-6-PD	Drug-induced hemolytic anemia
Lactate dehydrogenase (LD)	Myocardial infarction, hepatic disorders, carcionomas
Lipase (LPS)	Acute pancreatitis
Leucine aminopeptidase (LAP)	Hepatobiliary disorders
5'-Nucleotidase (5'NT)	Hepatobiliary disorders
Pseudocholineesterase (PChE) (butyrylcholine as substrate) ↓	Organophosphate poisoning
Ceruloplasmin (Copper-oxidase) ↓	Wilson's disease (abnormal Cu metabolism)

Thank you